

When Machine Learning Fails: Diagnosing and Repairing Shortcut Learning in Online Shoppers

Mini-Project Report

UCI Online Shoppers Purchasing Intention Dataset

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Abstract

This mini-project studies a deliberately induced failure mode in a non-linear classifier trained on the UCI Online Shoppers Purchasing Intention dataset. The model is intentionally forced to fail under a deployment-like temporal shift: it is first allowed to learn shortcut correlations from calendar and acquisition proxies, then evaluated on late-year sessions where those correlations are less reliable. The analysis follows a research structure: symptom characterization, falsifiable causal hypothesis, controlled intervention, targeted correction, validation with uncertainty, and threats to validity.

The central claim is deliberately modest: the experiments do not prove the real-world causes of online purchases. They show that the prediction function can become brittle when it relies on non-invariant shortcut variables, and that blocking this failure path improves robustness under the project protocol.

1. Research Question and Chosen Dataset

Dataset: UCI Online Shoppers Purchasing Intention, a binary classification task where the target is whether a web session ends in purchase.

Research question: Does a non-linear classifier trained on Online Shoppers sessions learn a spurious shortcut from calendar and acquisition proxies (**Month**, **TrafficType**, **VisitorType**, device/browser fields), such that it appears acceptable under an IID random split but loses robustness on late-year deployment months? If yes, can an invariant behavioral representation plus temporally adjacent validation improve out-of-period robustness without relying on that shortcut?

This question is falsifiable. The shortcut hypothesis would be weakened if:

- the shortcut-only model generalized as well as the behavioral model on Oct–Dec sessions;
- suspected shortcut variables showed no meaningful train-deployment drift;
- prediction-function interventions on shortcut variables did not change model behavior;
- permutation importance did not implicate shortcut variables.

2. Reference Model and Observed Symptom

The deliberately broken model is a Random Forest, satisfying the non-linear model-family constraint. It is restricted to shortcut-like variables: month, special-day proximity, operating system, browser, region, traffic type, visitor type, and weekend.

Two evaluation regimes are compared:

- **random split**, where seasonal regimes are mixed across train and test;
- **temporal deployment split**, where the model is trained on sessions up to September and evaluated on October–December.

The model was intentionally forced to fail under deployment-like temporal shift. Random validation leaks seasonal regimes across train and test, so it can overstate robustness. The

symptom is characterized using PR-AUC, purchase recall, balanced accuracy, expected calibration error, Brier score, confusion matrices, and subgroup metrics by month.

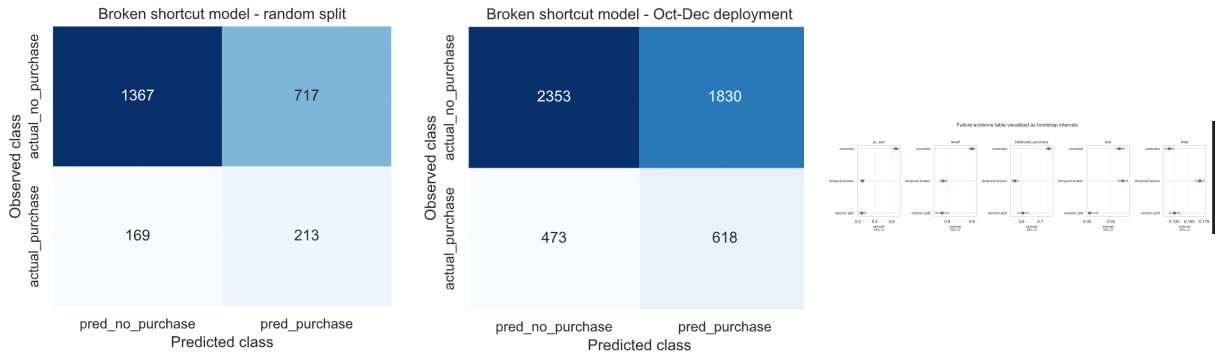


Figure 1: *Confusion matrices for the shortcut model under random validation and temporal deployment, alongside the failure evidence with bootstrap confidence intervals*

Examples of report-ready statements produced by the executed notebooks include:

- PR-AUC changed from the random-validation value to the temporal-deployment value.
- Purchase recall changed from the random-validation value to the temporal-deployment value.
- Expected calibration error changed under temporal deployment.
- The corrected model's late-year PR-AUC, recall, balanced accuracy, and calibration are reported on the same deployment months.

3. Causal Hypothesis and Controlled Experiment

Causal hypothesis: The model relies on non-causal calendar/acquisition shortcuts. These variables are correlated with purchase in the observed training distribution but are not invariant causes of purchase behavior. When the temporal regime changes, the shortcut breaks.

The controlled experiment manipulates the representation while holding the model family, target, split logic, and metrics fixed:

- **treatment expected to fail:** shortcut-only representation;
- **positive control:** behavioral-intent representation;
- **leakage sensitivity control:** behavioral representation without [PageValues](#);
- **negative control:** corrected representation excludes [Month](#), so the shortcut path is blocked by design.

The intervention is causal with respect to the prediction function: the experiment changes what information the fitted model can use, not the real-world shopping process.

The causal evidence is triangulated:

- drift analysis verifies that suspected shortcut variables change between train and deployment regimes;
- controlled representation tests compare shortcut-only and behavioral models under the same temporal split;

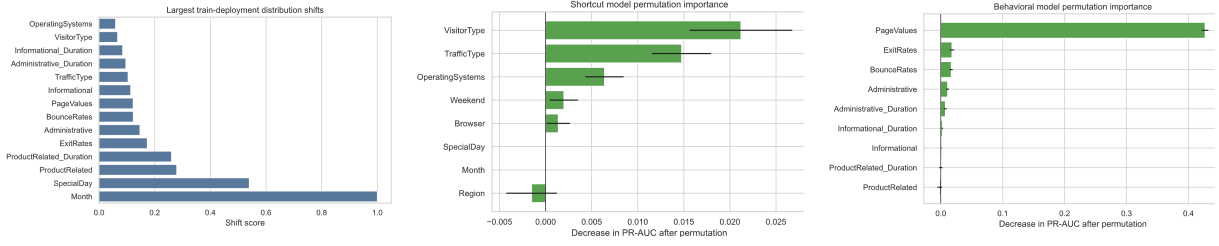


Figure 2: Train-deployment drift scores and permutation importance analyses for shortcut and behavioral features.

- prediction-function interventions permute suspected variables while holding the trained model fixed;
- permutation importance checks whether the fitted model depends on shortcut variables;
- SHAP is included as an interpretability supplement, not as the main causal proof.

4. Proposed Correction and Evaluation

The correction targets the cause rather than merely masking the symptom:

- remove unstable calendar/acquisition proxy variables;
- train on behavioral intent features;
- select the threshold on September, a temporally adjacent validation month;
- use cost-aware class weighting and a recall-at-minimum-precision objective;
- report bootstrap uncertainty and seed variance.

This is not a generic SMOTE-style correction. The repair is tied to the failure mechanism: if shortcut reliance is the cause, blocking those variables and selecting decisions under a closer temporal regime should improve deployment robustness.

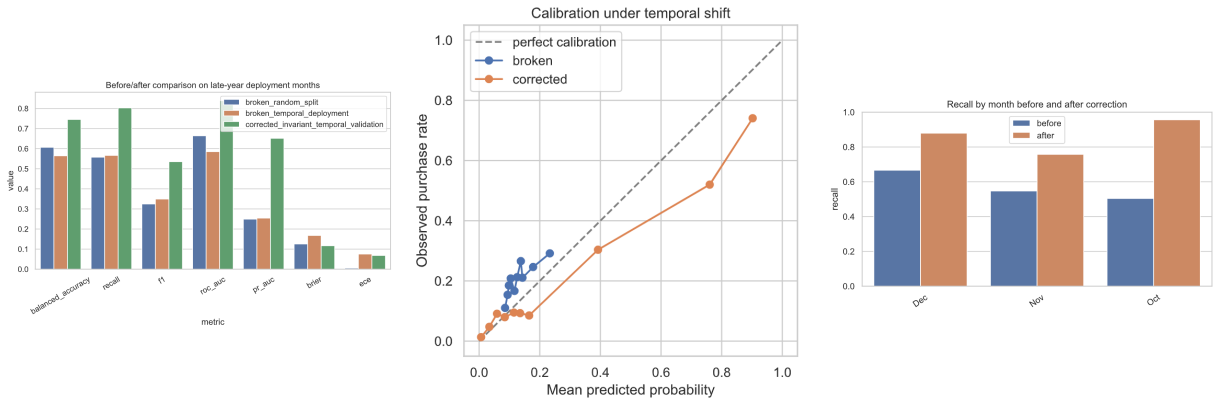


Figure 3: Before/after evaluation of the corrected model, including deployment performance, calibration behavior, and subgroup recall robustness.

The central quantitative deliverable is the Failure Evidence Table (`table08_failure_evidence_bootstrap_ci.csv`). It includes random split, temporal split, and corrected model estimates with 95% bootstrap confidence intervals for PR-AUC, recall, balanced accuracy, ECE, and Brier score.

	model	metric	estimate	ci_low	ci_high	threshold	n	n_boot
0	broken_random_split	pr_auc	0.2492	0.2205	0.2938	0.1525	2466	500
1	broken_random_split	recall	0.5576	0.5063	0.6068	0.1525	2466	500
2	broken_random_split	balanced_accuracy	0.6068	0.5801	0.6348	0.1525	2466	500
3	broken_random_split	ece	0.0053	0.0036	0.0223	0.1525	2466	500
4	broken_random_split	brier	0.1257	0.1172	0.1344	0.1525	2466	500
5	broken_temporal_deployment	pr_auc	0.2544	0.2373	0.2748	0.1230	5274	500
6	broken_temporal_deployment	recall	0.5665	0.5403	0.5968	0.1230	5274	500
7	broken_temporal_deployment	balanced_accuracy	0.5645	0.5508	0.5817	0.1230	5274	500
8	broken_temporal_deployment	ece	0.0755	0.0655	0.0856	0.1230	5274	500
9	broken_temporal_deployment	brier	0.1679	0.1609	0.1748	0.1230	5274	500

Figure 4: Tabele of failure evidence bootstrap ci

5. Bonus Failure Mode: Class Imbalance Hiding Purchase Failure

Symptom: Purchases are the minority class, so a default threshold can produce superficially acceptable accuracy while failing to detect purchase sessions.

Hypothesis: The default decision threshold privileges majority-class behavior more than purchase detection.

Controlled test: Keep the corrected model and predicted probabilities fixed; manipulate only the threshold rule.

Correction: Choose a threshold on validation data with an explicit recall objective subject to a minimum precision constraint.

This failure mode is complementary to the main one. Representation repair addresses shortcut learning; threshold repair addresses asymmetric decision costs. The results are written to:

- `outputs/metrics/table12_bonus_threshold_failure.csv`;
- `outputs/metrics/table13_bonus_threshold_bootstrap_deltas.csv`.

	metric	delta_after_minus_before	ci_low	ci_high	n_boot
1	accuracy	-0.1374668185419028	-0.14770572620401967	-0.12693875616230565	500
2	recall	0.08340971585701185	0.0677243989329258	0.10166569257744011	500
3	precision	-0.2142197300551752	-0.23057826144302782	-0.19572500519192967	500
4	f1	-0.12827847608923548	-0.14301347207849788	-0.11236191968031804	500
5	balanced_accuracy	-0.055832794474075964	-0.06593132949964581	-0.04515255351046507	500

	model	n	prevalence	threshold	accuracy	balanced_accuracy	precision	recall	f1	roc_auc	pr_auc	brier	ece
0	corrected_default_0.50	5274	0.2069	0.5000	0.8491	0.8012	0.6157	0.7195	0.6636	0.8389	0.6516	0.1168	0.0684
1	corrected_cost_aware_threshold	5274	0.2069	0.1426	0.7116	0.7454	0.4015	0.8029	0.5353	0.8389	0.6516	0.1168	0.0684

Figure 5: Bonus analysis of threshold failure and bootstrap confidence interval deltas.

6. Threats to Validity

Observational-data limitation: The experiments manipulate model inputs and validation protocols, not real user behavior. The causal claim is about the prediction function, not about the true causal drivers of purchase.

Coarse temporal axis: `Month` is not a true timestamp. The dataset is not a multi-year production log, so Oct–Dec deployment is a plausible stress test rather than a complete production replay.

Confounding between seasonality and campaigns: `Month`, `TrafficType`, and

VisitorType may reflect promotions, traffic acquisition strategies, or external demand shocks. The project treats them as unstable proxies, not as intrinsically invalid variables.

Leakage risk: **PageValues** may encode in-session information close to the purchase event. The no-PageValues control checks whether the correction depends entirely on this potentially leaky signal.

Threshold dependence: The cost-aware threshold is linked to an explicit preference for purchase recall under a minimum precision constraint. Different business costs would justify a different threshold.

Statistical uncertainty: Bootstrap intervals and seed variance reduce the risk of overinterpreting a single run, but they do not eliminate uncertainty. Some month-level subgroups remain small.

External validity: The conclusion should be retested on newer traffic logs before deployment. The current project demonstrates a failure mechanism, not a universal guarantee.

Model-family dependence: The analysis uses a tree ensemble. A neural network or gradient boosting model might learn a related but not identical shortcut profile.

7. Conclusion and What We Learned

The project demonstrates the intended scientific workflow:

- a model was intentionally forced to fail;
- the failure was made visible with metrics, confidence intervals, confusion matrices, subgroup analysis, calibration, and drift diagnostics;
- the proposed cause was tested through controlled representation changes and prediction-function interventions;
- the correction targeted the identified mechanism rather than applying a generic metric hack;
- uncertainty, leakage risk, and external validity limits were discussed explicitly.

The corrected model is not presented as perfect. The stronger and more defensible conclusion is that the original model failed for an experimentally diagnosable reason, and that a targeted intervention reduced that failure under a deployment-like temporal protocol.